

EEE 308L

Electronic Circuits I Laboratory

Lab Report



Inspiring Excellence

Section:06 , Group No:04 , Experiment no:01

Name of the experiment: *Study of Operational Amplifier as Amplifier*

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Name of the Experiment: Study of Operational Amplifier as Amplifier.

Objective: To study op-amp as an amplifier in both inverting and non-inverting configuration.

Equipment:

- | | |
|-------------------------|----------|
| ▪ IC: 741 | 1 Piece |
| ▪ Resistor: 10k | 2 Pieces |
| ▪ Resistor: 100k | 1 Piece |
| ▪ AT-7000 Trainer Board | 1 Unit |
| ▪ DC Power Supply | 1 Unit |
| +15 volts, -15 Volts | |
| ▪ Function Generator | 1 Unit |
| ▪ Oscilloscope | 1 Unit |
| ▪ Multimeter | 1 Unit |
| ▪ Wires | |

Circuit Diagram:

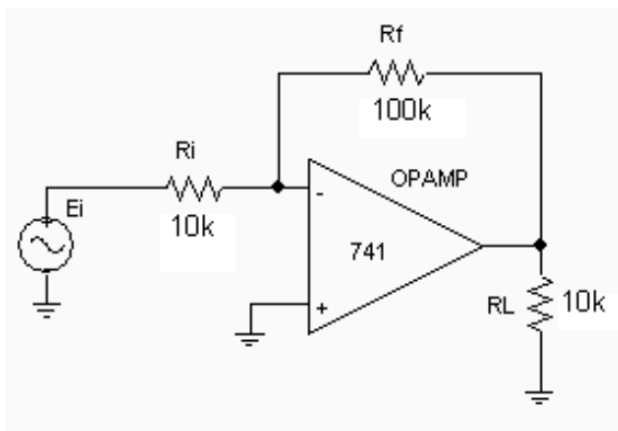


Fig-1: Inverting Amplifier

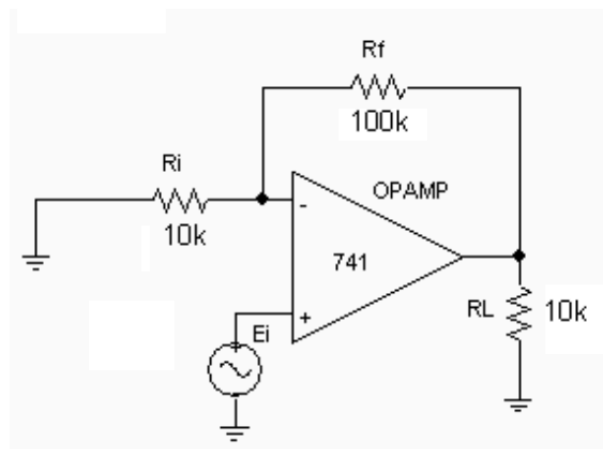


Fig-2: Non-Inverting Amplifier

Experimental Setup:

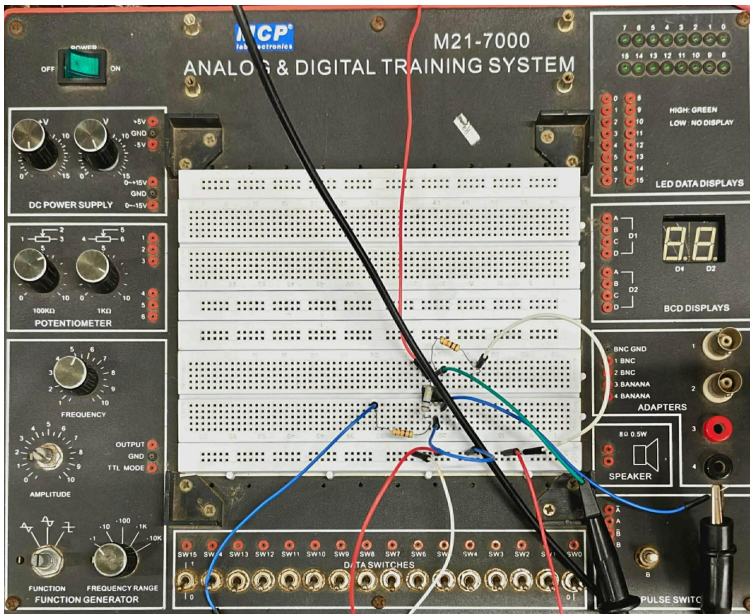


Fig-3:Inverting Circuit

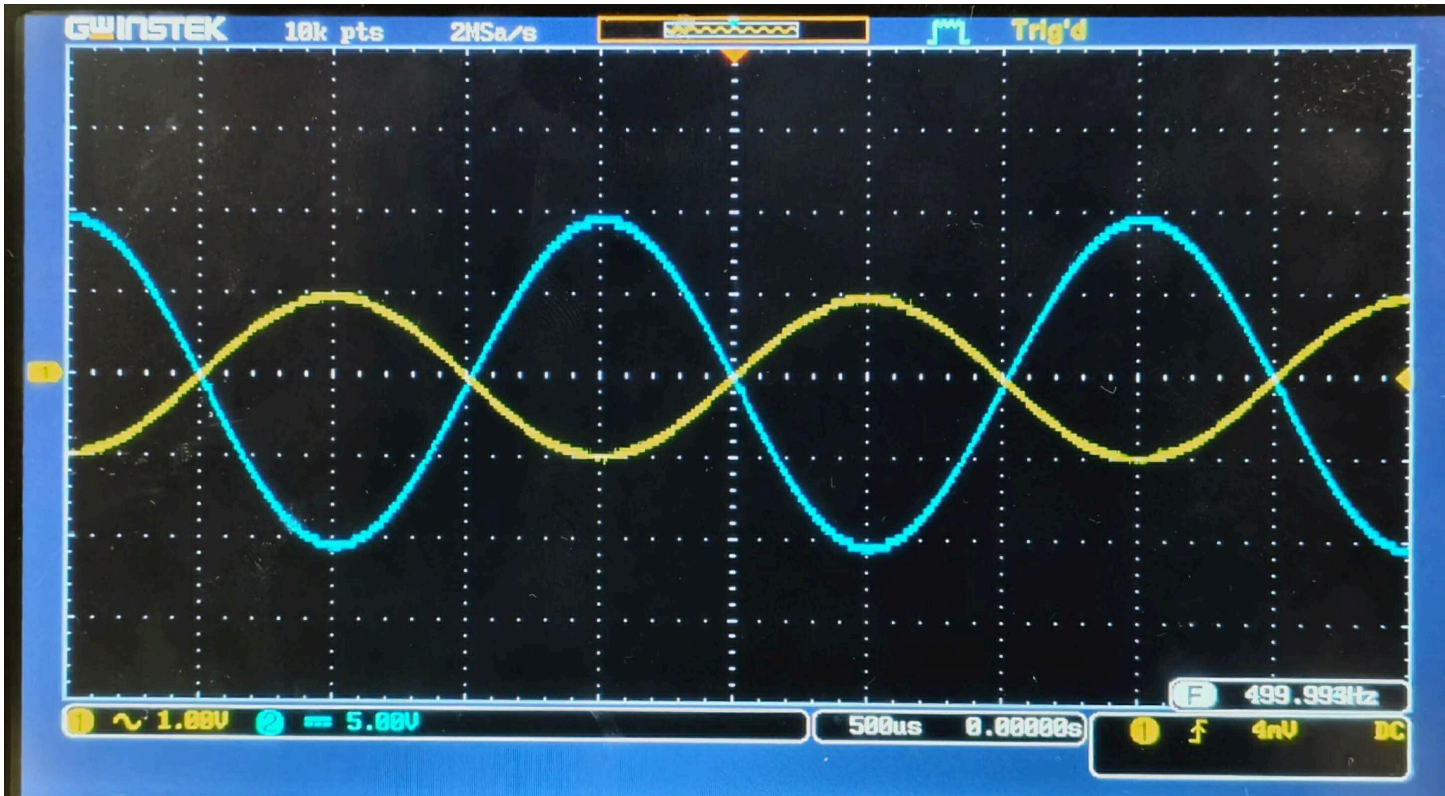


Fig-4:Output of Inverting Signal

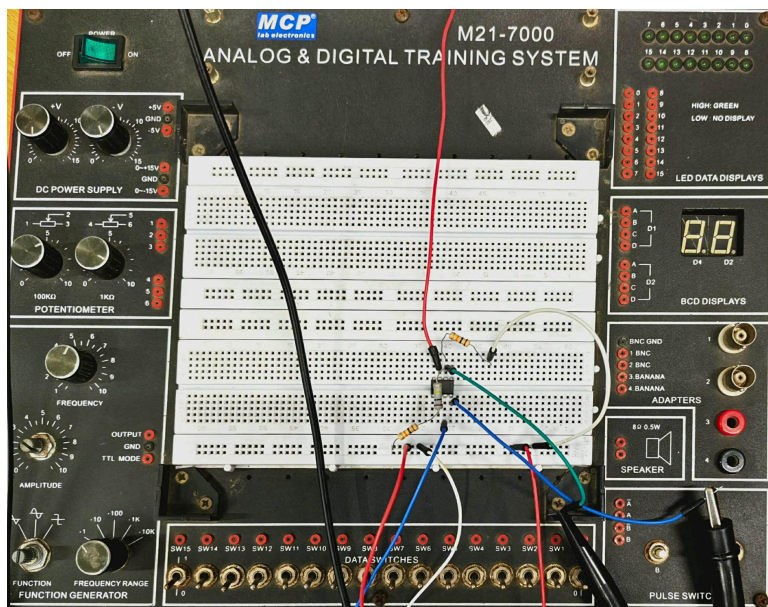


Fig-5:Non-Inverting Circuit



Fig-6:Output of Non Inverting Signal

Data:

Data Sheet

Group-4
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23221042
23221101
24121205
24121307

☒ Inverting Amplifier:

R_i (k Ω)	R_f (k Ω)	$A_{CL} = -R_f/R_i$	E_i (pk-pk) (V)	V_o (pk-pk) (V)	Gain $A_o = V_o/E_i$
9.85	101.2	-10.27	2.02	20.04	9.92

Calculation:

$$A_{CL} = -R_f/R_i$$

$$= -\frac{101.2}{9.85}$$

$$= -10.27$$

$$\text{Gain, } A_o = \frac{V_o}{E_i}$$

$$= \frac{20.04}{2.02}$$

$$= 9.92$$

☒ Non-inverting Amplifier:

R_i (k Ω)	R_f (k Ω)	$A_{CL} = 1 + R_f/R_i$	E_i (pk-pk) (V)	V_o (pk-pk) (V)	Gain $A_o = V_o/E_i$
9.85	101.2	11.27	2.10	22.5	10.71

Calculation:

$$A_{CL} = 1 + R_f/R_i$$

$$= 1 + \frac{101.2}{9.85} = 11.27$$

$$A_o = \frac{V_o}{E_i} = \frac{22.5}{2.10}$$

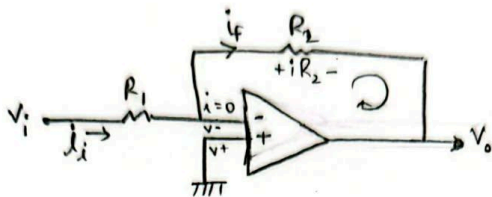
$$= 10.71$$

~~2021~~
19/02/2026

Question:

1. Derive the following closed-loop gain of the inverting and non-inverting amplifier of op amp $A_{cl} = -R_f/R_i$ and $A_{cl} = 1 + R_f/R_i$

Inverting Amplifier: Output Voltage and Gain



$$R_{id} = \infty \quad \text{so, } i = 0$$

and

$$A_{ol} = \infty \quad \text{so, } V^+ = V^- \text{ [virtual short circuit]}$$

Since,

$$V^+ = 0 \quad \text{so } V^- = 0$$

Now,

$$i_i = \frac{V_i - V^-}{R_1} = \frac{V_i}{R_1} \rightarrow \textcircled{i}$$

KVL in the loop,

$$-i_f R_2 - V_o + V^- = 0$$

$$\Rightarrow V_o = -i_f R_2 + V^-$$

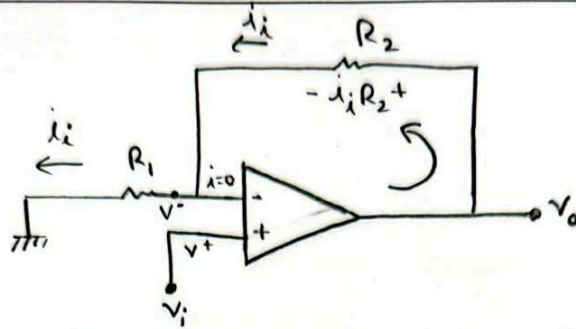
$$\Rightarrow V_o = -i_i R_2 \quad [i_f = i_i \text{ and } V^- = 0]$$

$$\Rightarrow V_o = -\frac{V_i}{R_1} R_2$$

$$\Rightarrow \frac{V_o}{V_i} = -\frac{R_2}{R_1}$$

$$\boxed{A_{cl} = -\frac{R_2}{R_1}} \rightarrow \text{for inverting}$$

Non-Inverting Amplifier: Output Voltage and Gain



for ideal case, $R_{id} = \infty$, so $i_i = 0$
and $A_{ol} = \infty$ so, $v^+ = v^-$

Now,

$$v^- = v_i = v^+ \\ i_i = \frac{v^- - 0}{R_1} = \frac{v^-}{R_1} = \frac{v_i}{R_1} \rightarrow \textcircled{i}$$

KVL in the loop,

$$\begin{aligned} v_o - i_i R_2 - v^- &= 0 \\ \Rightarrow v_o &= i_i R_2 + v^- \\ &= \frac{v_i}{R_1} R_2 + v_i \\ &= v_i \left(\frac{R_2}{R_1} + 1 \right) \\ \Rightarrow \frac{v_o}{v_i} &= 1 + \frac{R_2}{R_1} \end{aligned}$$

$$\boxed{A_{cl} = 1 + \frac{R_2}{R_1}} \rightarrow \text{for non inverting}$$

2. Between the two types of amplifiers which one will you prefer and why?

ANS: Between inverting and non-inverting op-amps I will prefer the inverting amplifier for the following reasons-

- If we calculate the difference of the actual gain and theoretical gain for both cases, the inverting circuit has the least difference between the actual value and theoretical value.
- For the equations we have derived for both cases, we can see that the A_{ct} for non-inverting will always be greater than 1. So, if we try to design an amplifier circuit for scaling down the output circuit, we can not do it in this configuration. On the other hand, inverting circuits do not have limitations. Although, we do not usually scale down our output signal but this is a thing we can consider for exceptional uses.
- Inverting the amplifier circuit gives some distortion to the output which may not be seen but can harm in larger implementations.

Conclusion:

In this experiment, the performance of the IC 741 op-amp was analyzed in both inverting and non-inverting configurations. For the inverting amplifier, the measured gain was approximately 10, which closely matched the theoretical value $A_v = -R_f/R_i$, and the output showed a 180° phase shift as expected. For the non-inverting amplifier, the measured gain was about 10.6, which was close to the theoretical value $A_v = 1 + R_f/R_i$, and the output remained in phase with the input signal.

The small differences between theoretical and measured values may be due to component tolerances, internal characteristics of the IC 741, or minor circuit losses. Overall, the experiment successfully verified the working principles and gain behavior of inverting and non-inverting op-amp amplifiers using negative feedback.

